

Latent Arithmetic Measures the Origin: A No-Retraining Intervention on Audio Autoencoders

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Abstract—Audio latents are increasingly judged by arithmetic: interpolate two codes and decode a mix, subtract a stem’s code and decode the residual. One of these probes is not a property of the autoencoder. Shifting the latent origin by the encoded silence $f(0)$ leaves the autoencoding function and every unit-sum affine combination exactly unchanged, for arbitrary nonlinear encoders and decoders, yet it moves stem-subtraction scores by decibels. On five fixed checkpoints and a pre-registered held-out corpus of 240 recordings, the shift reduces the apparent advantage of mixing supervision by 87–95%, leaving 0.12 to 0.42 dB. Convex mixing, which the same shift provably cannot touch, improves by +1.7 and +1.3 dB at two compression rates. The two probes therefore support different conclusions about the same checkpoints, and latent arithmetic should be reported against a declared origin.

Index Terms—representation learning, equivariant representations, audio autoencoders, latent arithmetic, evaluation methodology

I. INTRODUCTION

Audio mixing is linear on waveforms: $\text{mix}(x_1, x_2, \alpha) = \alpha x_1 + (1-\alpha)x_2$. This makes latent arithmetic an attractive interface to a learned audio representation. For an encoder f and decoder g , a latent space is *mixing-equivariant* when

$$g(\alpha f(x_1) + (1-\alpha)f(x_2)) \approx \alpha x_1 + (1-\alpha)x_2, \quad (1)$$

and a natural companion probe is stem removal by subtraction, $g(f(x_{\text{mix}}) - f(x_{\text{stem}})) \approx x_{\text{res}}$. Where these hold, editing and composition happen in the latent domain without a dedicated separation or synthesis model, a prerequisite for latent-diffusion systems [1], [2] that compose independently modeled sources. Standard autoencoders do not satisfy Eq. 1: their objectives reward reconstruction fidelity, not algebraic structure, and in quantized codecs (DAC [3], EnCodec [4]) an interpolated point is not a codebook entry. Torres et al. [5] induce the property in a consistency autoencoder by decoding averaged latents against the true mix.

This paper asks a prior question: *what do these arithmetic scores measure?* The two probes differ in one respect that is easy to overlook. The coefficients in Eq. 1 sum to one; those of subtraction sum to zero. Latents are defined only up to a choice of origin, and a shift of that origin cancels in the first case but not in the second. Section III makes this precise for arbitrary nonlinear f and g : recentering the latent space

at $f(0)$ leaves the autoencoding function and every unit-sum combination pointwise identical, while mapping subtraction to $g(f(x_{\text{mix}}) - f(x_{\text{stem}}) + f(0))$. Any change a subtraction score shows under this map is therefore a property of the coordinate system, not of the model as a function.

We run that intervention on five fixed checkpoints of a waveform GAN autoencoder, spanning two compression rates and three training objectives, with no retraining and no change to the evaluation audio. Our contributions:

- 1) A reparameterization argument (Section III) showing that subtraction-style latent arithmetic is coordinate-dependent while reconstruction and convex mixing are not, and identifying the model’s own reconstruction of the target as the coordinate-invariant reference.
- 2) The intervention applied to fixed checkpoints (Section V-A), on the test set and on a pre-registered held-out corpus: recentering reduces the advantage that mixing supervision appears to have on stem subtraction by 87–95%, and shrinks the between-model gap on 229–236 of 240 held-out recordings, leaving a residue below half a decibel.
- 3) Evidence that the two probes disagree on the same checkpoints (Section V-B): on convex mixing, which the intervention cannot affect, the supervision delivers a measurable gain at both compression rates and reduces level error, not only spectral error.
- 4) A methodological correction (Section V-C): the widely used decode-vs-decode SI-SDR is inflated by an adversarial term and by decoder sampling noise; we report a ground-truth-referenced variant.

II. RELATED WORK

Neural audio codecs. DAC [3], EnCodec [4] and SoundStream [6] pair convolutional encoders with HiFi-GAN decoders [7] and multi-scale discriminators, optimizing reconstruction without latent structure; their residual vector quantization also precludes latent interpolation. RAVE [8] learns a variational latent space but does not target mixing equivariance, and disentanglement regularizers [9] encourage generic smoothness rather than a specific algebraic identity. Mixing-equivariance follows the symmetry-based view of representation learning [10], though convex mixing is an affine com-

bination, not a group action. **Linear audio representations.** Music2Latent [11] is a consistency autoencoder with partial linearity as a byproduct. Torres et al. [5] make it approximately linear by conditioning the decoder on the average of separately encoded latents while denoising the true mix under the unchanged consistency loss, a principle related to mixup [12]. Their additivity and separation scores compare two decodes, a protocol Section V-C shows to be confounded. No prior work reports latent arithmetic against a declared origin.

III. THE LATENT ORIGIN IS A FREE COORDINATE

Proposition. Let $f : \mathcal{X} \rightarrow \mathcal{Z}$ and $g : \mathcal{Z} \rightarrow \mathcal{X}$ be arbitrary maps, and define the recentered pair

$$f_c(x) = f(x) - f(0), \quad g_c(z) = g(z + f(0)). \quad (2)$$

Then (i) $g_c(f_c(x)) = g(f(x))$ for every x ; (ii) for any $x_1, \dots, x_n$ and any coefficients with $\sum_i \alpha_i = 1$, $g_c(\sum_i \alpha_i f_c(x_i)) = g(\sum_i \alpha_i f(x_i))$; and (iii) for the coefficients $(1, -1)$,

$$g_c(f_c(x) - f_c(y)) = g(f(x) - f(y) + f(0)). \quad (3)$$

Proof. $f_c(x) + f(0) = f(x)$ gives (i). For (ii), $\sum_i \alpha_i f_c(x_i) = \sum_i \alpha_i f(x_i) - f(0) \sum_i \alpha_i = \sum_i \alpha_i f(x_i) - f(0)$, and g_c adds $f(0)$ back. For (iii) the coefficients sum to zero, so the offset cancels in the latent difference, $f_c(x) - f_c(y) = f(x) - f(y)$, and g_c adds $f(0)$ inside g . $\square$

Three consequences shape the rest of the paper. First, (f_c, g_c) is the same autoencoder as (f, g) as far as reconstruction and Eq. 1 are concerned: by (i) and (ii) those quantities are pointwise identical, for arbitrary nonlinear networks, with no retraining and no approximation. Second, by (iii) subtraction is not: recentering shifts it by one encode of silence, and Section V-A measures a several-decibel shift. A model ranking obtained from raw subtraction can therefore be altered by a change of coordinates that provably leaves reconstruction and mixing untouched, so such a ranking is not by itself a statement about the representation. Third, when f is affine, $f(x) = Ax + b$, we have $f(x) - f(y) + f(0) = f(x - y)$ and Eq. 3 decodes exactly $g(f(x_{\text{res}}))$, the model’s own reconstruction of the residual, whatever the decoder does. That encode–decode reconstruction is the reference we report against; the shortfall from it is the *linearity tax*. Real encoders depart from affine, so the latent recovery is exact only in that limit.

IV. EXPERIMENTAL SETUP

A. Models and supervision

The primary model is a waveform autoencoder (DAC-style 1-D convolutional encoder, HiFi-GAN decoder [7], combined multi-scale-STFT and multi-period discriminator) at 44.1 kHz on 1-second mono chunks, trained on a ~ 950 -hour union of FMA-large, MAESTRO and MUSDB18-HQ [13].¹ Two

¹Code, training configurations, pretrained weights, every metric reported here, and audio examples: <https://github.com/nurdauletakhanov/musicgen>

compression regimes are used: $7.66 \times$ ($d=128$) and $15.3 \times$ ($d=64$).

Mixing supervision is the explicit-loss form of the additivity principle of [5]. We form pairs (x_i, x_j) with a fixed-point-free permutation and draw $\alpha \sim U[0, 1]$; with $\bar{x} = \alpha x_i + (1 - \alpha)x_j$ and $\bar{z} = \alpha f(x_i) + (1 - \alpha)f(x_j)$, the decode-mixing loss is, as implemented,

$$\mathcal{L}_{\text{mix}}^{\text{dec}} = \mathcal{L}_{\text{recon}}(g(\bar{z}), \bar{x}) + \|g(\bar{z}) - \bar{x}\|_1, \quad (4)$$

where $\mathcal{L}_{\text{recon}}$ is the weighted multi-resolution STFT and mel loss also used for reconstruction; the mixed path carries the extra unweighted waveform ℓ_1 term, the ordinary reconstruction path does not. Gradients flow through g and into f via $\bar{z}$. The total objective is $\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{GAN}} + \lambda \mathcal{L}_{\text{mix}}^{\text{dec}} + \lambda_{\ell_2} \|z\|^2$; a variant additionally applies the discriminator to the mixed path (“+disc”). Mixing losses are added either as a 25k-step fine-tune over a converged baseline ($7.66 \times$) or from scratch over 250k steps ($15.3 \times$), each against a matched control that is identical except that mixing is disabled. All runs use a single NVIDIA RTX 5090.

B. Metrics

We report reconstruction $\text{SI-SDR}_{\text{rec}} = \text{SISDR}(g(f(x)), x)$ and, at $\alpha=0.5$, two equivariance views: $\text{SI-SDR}_{\text{lin}} = \text{SISDR}(g(\bar{z}), g(f(\bar{x})))$ (decode-vs-decode, relative to the model’s own reconstruction of the mix) and $\text{SI-SDR}_{\text{lin}}^{\text{gt}} = \text{SISDR}(g(\bar{z}), \bar{x})$ (vs. ground truth, the quantity in Eq. 1 up to scale and comparable across models). We also report $\ell_{\text{lat}} = \|\bar{z} - f(\bar{x})\|^2 / \|f(\bar{x})\|^2$ and $\text{MixRate} = \mathcal{L}_{\text{mix}}(g(\bar{z}), \bar{x}) / \mathcal{L}_{\text{mix}}(g(f(\bar{x})), \bar{x})$, with $\mathcal{L}_{\text{mix}}$ the three-term mixed-path loss of Eq. 4 in both numerator and denominator (< 1 beats the encode-then-decode path). $\text{SI-SDR}_{\text{lin}}$ shares the decode-vs-decode structure of prior additivity metrics [5] and can move without ground-truth equivariance moving (Section V-B), so $\text{SI-SDR}_{\text{lin}}^{\text{gt}}$ and MixRate are our metrics of record. SI-SDR fits a gain before scoring, so we report that gain separately, on 3,000 samples. FAD uses music-finetuned LAION-CLAP embeddings of non-overlapping 10s windows at 48 kHz, not VGGish, so values are not comparable with VGGish FAD.

C. Protocol and what the intervals cover

Subtraction uses 49 of the 50 MUSDB18 test tracks (one is excluded because its stems sum to the mixture at only 18.5 dB), giving 5,880 (track, stem, chunk) units; chunk selection is a pure function of the seed and the sorted track list, so all models see identical units and every contrast is paired. Confidence intervals come from a paired cluster bootstrap over recordings (10^4 resamples), the correlated unit: 49 tracks on the test set and 240 on the held-out corpus. Mixing pairs join two chunks, about 80% from the same recording, so those intervals weight a pair by the multiplicity of each member. Intervals are a *conditional diagnostic*: they cover evaluation variance over audio for these checkpoints, and cover neither training-seed variance nor checkpoint selection. For the held-out subtraction only, a unit is scored where the removed stem

Model	drums	bass	vocals	other	all	gap↓
7.66× no mix	+4.2	+0.4	+5.2	+4.0	+3.4	5.2
+ $f(0)$	+8.8	+6.8	+9.6	+8.0	+8.3	0.4
7.66× + $\mathcal{L}_{\text{dec}}$	+5.7	+4.9	+7.3	+5.2	+5.8	2.9
+ $f(0)$	+8.5	+6.9	+9.4	+8.3	+8.3	0.5
7.66× + $\mathcal{L}_{\text{dec}}$ +disc	+6.2	+5.4	+7.5	+5.2	+6.1	2.6
+ $f(0)$	+8.5	+7.0	+9.3	+8.1	+8.2	0.4
15.3× no mix	+1.2	-1.6	+3.0	+2.2	+1.2	6.6
+ $f(0)$	+7.3	+5.8	+8.8	+7.2	+7.3	0.5
15.3× + $\mathcal{L}_{\text{dec}}$ +disc	+5.4	+4.7	+6.9	+4.0	+5.2	2.4
+ $f(0)$	+7.3	+6.1	+8.5	+7.2	+7.3	0.4
M2L +mix	-9.9	-12.3	-9.9	-12.7	-11.2	8.4

TABLE I

STEM REMOVAL BY LATENT SUBTRACTION (SI-SDR, dB; MUSDB18 TEST, 5,880 UNITS FROM 49 TRACKS). “+ $f(0)$ ” ROWS READ THE *same* CHECKPOINT IN RECENTERED COORDINATES. “GAP” IS THE LINEARITY TAX AGAINST EACH MODEL’S OWN ENCODE-DECODE RECONSTRUCTION OF THE RESIDUAL (LOWER IS BETTER).

is active (24,648 of 28,800); this excludes silent *targets* and does not make the SI-SDR reference nonsilent, that reference being the remaining mixture.

That selection needs disclosure. The released weights are the checkpoint with the lowest validation loss, and the validation loader read the test split. For every 7.66× run this coincides with the final step, so selection did not alter those weights; both 15.3× runs were selected at step 240k of 250k. Contrasts are between matched configurations under identical selection, and the origin intervention is applied to fixed checkpoints so it is unaffected, but absolute generalization claims should be read with this in mind. The code now supports a held-out validation split.

V. RESULTS

A. Recentering removes the subtraction advantage

We remove a stem by latent subtraction, $\hat{x} = g(f(x_{\text{mix}}) - f(x_{\text{stem}}))$, and score SI-SDR against the true residual. This is an *oracle* setting: the removed stem is available and encoded, so it probes the representation’s arithmetic, not blind separation; its use is latent-domain editing when stem codes exist. The intervention of Section III replaces this by $g(f(x_{\text{mix}}) - f(x_{\text{stem}}) + f(0))$, which by Eq. 3 is the same model read in recentered coordinates.

In raw coordinates the supervision looks decisive. It raises subtraction quality from +3.4 to +5.8 dB at 7.66× and from +1.2 to +5.2 at 15.3× (Table I).

Recentering removes most of it. Adding $f(0)$ gains +2.0 to +6.0 dB for every configuration and leaves each within 0.4–0.5 dB of its own reconstruction reference; the models that gain most looked worst. The between-model advantage falls to -0.03, -0.07 and +0.01 dB, all three intervals containing zero (Table II), and the reduction holds on 49 of 49 tracks in every pair (Fig. 1, right). The no-mixing models also carry the larger offset ($\|f(0)\|$ 6.6 versus 5.4 at 7.66×, 18.1 versus 14.8 at 15.3×), though offset norm alone does not determine the

Treatment vs. its control	raw	recent.	95% CI	tracks
<i>MUSDB18 test (49 tracks)</i>				
$\mathcal{L}_{\text{mix}}^{\text{dec}}, 7.66\times$	+2.33	-0.03	[-0.15, +0.10]	49/49
$\mathcal{L}_{\text{mix}}^{\text{dec}} + \text{disc}, 7.66\times$	+2.64	-0.07	[-0.19, +0.07]	49/49
$\mathcal{L}_{\text{mix}}^{\text{dec}} + \text{disc}, 15.3\times$	+4.02	+0.01	[-0.19, +0.21]	49/49
<i>MoisesDB, held out (240 tracks)</i>				
$\mathcal{L}_{\text{mix}}^{\text{dec}}, 7.66\times$	+2.27	+0.29	[+0.20, +0.39]	229/240
$\mathcal{L}_{\text{mix}}^{\text{dec}} + \text{disc}, 7.66\times$	+2.47	+0.12	[+0.03, +0.22]	231/240
$\mathcal{L}_{\text{mix}}^{\text{dec}} + \text{disc}, 15.3\times$	+4.50	+0.42	[+0.29, +0.55]	236/240

TABLE II

THE INTERVENTION ON FIXED CHECKPOINTS. ADVANTAGE IN STEM-SUBTRACTION SI-SDR (dB) OF EACH MIXING-TRAINED MODEL OVER THE MATCHED CONTROL AT THE SAME COMPRESSION RATE TRAINED WITHOUT MIXING SUPERVISION, BEFORE AND AFTER RECENTERING, WITH THE PAIRED CHANGE AND THE NUMBER OF TRACKS ON WHICH THE ADVANTAGE SHRINKS. RECENTERING IS A CHANGE OF COORDINATES ONLY: NO RETRAINING, SAME UNITS, SAME AUDIO.

decoded error. **Held-out replication, and a residue.** Because these checkpoints were selected on the split we report, we repeat the intervention on MoisesDB [14], 240 recordings absent from training and from selection, under a protocol registered before the audio was downloaded, with frozen weights and verified hashes. Raw subtraction again favours the mixing models by +2.3 to +4.5 dB and recentering removes 87–95% of that, shrinking the gap on 229, 231 and 236 of the 240 recordings (Table II); on the rest it overshoots and reverses the sign. The residue is resolvable here, +0.29, +0.12 and +0.42 dB with intervals excluding zero, where the 49-track set could not. The three contrasts are $\mathcal{L}_{\text{mix}}^{\text{dec}}$ and $\mathcal{L}_{\text{mix}}^{\text{dec}} + \text{disc}$ against no mixing at 7.66× and the combined recipe against no mixing at 15.3×; the comparator figures below return to the MUSDB18 test set.

What survives. Per stem the recentered contrast is nonzero in both directions, +0.35 dB on bass against -0.34 on vocals at 15.3×.

Baselines without latent arithmetic. The unchanged mixture scores +17.1 dB against the residual, above every model’s reconstruction reference, because the removed stem is a small share of the energy: the metric certifies arithmetic relative to that reference, not removal. Recentered latent subtraction beats decode-then-subtract by +0.7 to +0.9 dB; raw subtraction loses to it everywhere.

B. Convex mixing tells a different story

By part (ii) of the proposition, recentering cannot change any quantity in Eq. 1. The mixing probe therefore reports on the autoencoder itself, and there the supervision does something (Table III). On the held-out corpus, with intervals from a dyadic recording bootstrap (each mixing unit involves two recordings), $\mathcal{L}_{\text{mix}}^{\text{dec}}$ improves ground-truth equivariance by +1.73 dB [+1.69, +1.78] at 7.66× and the combined recipe by +1.33 [+1.27, +1.39] at 15.3×, at a reconstruction change of +0.04 and -0.07 dB, and the adversarial term costs 0.49 dB [-0.51, -0.48] against $\mathcal{L}_{\text{mix}}^{\text{dec}}$ alone while raising

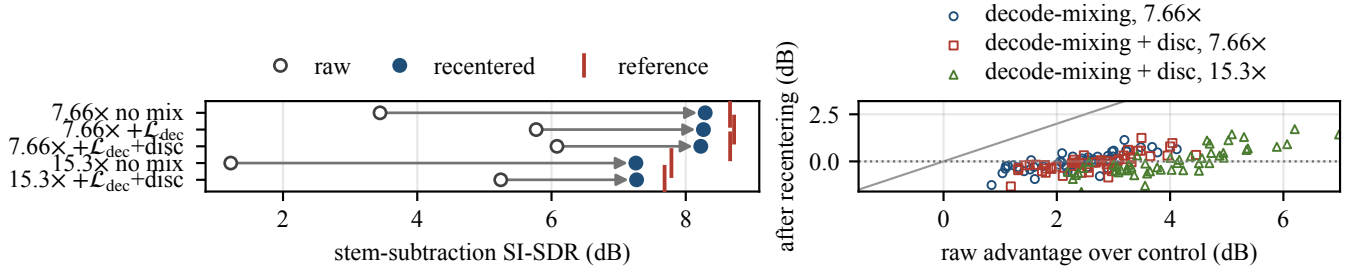


Fig. 1. The origin intervention on five fixed checkpoints. *Left*: stem-subtraction SI-SDR moves from raw coordinates to recentered ones (arrow), landing at each model’s own reconstruction of the residual (red tick); the raw ordering does not survive. *Right*: per-track advantage over the matched no-mixing control, raw (horizontal) against recentered (vertical), one point per track. Every point falls below the diagonal and the recentered values sit at zero.

Model	SDR _{rec}	SDR _{lin}	SDR _{lin} ^{gt}	MixRate	FAD
GAN, 7.66×, no mix	+11.3	+12.0	+8.0	1.203	0.045
GAN, 7.66×, + $\mathcal{L}_{dec}$	+11.4	+12.1	+10.2	0.972	0.044
GAN, 15.3×, no mix	+10.0	+10.6	+6.8	1.187	0.052
GAN, 15.3×, + $\mathcal{L}_{dec}$ +disc	+9.9	+14.6	+8.6	1.046	0.061
M2L, no mix (baseline)	-3.3	+5.0	-9.6	1.101	0.127
M2L, fine-tune control	-3.1	+5.1	-9.5	1.125	0.134
M2L, +mix	-2.6	+6.9	-7.8	1.086	0.119

TABLE III

CONVEX MIXING ACROSS COMPRESSION AND ARCHITECTURE (FULL TEST SET, $\alpha=0.5$, EMA WEIGHTS FOR MUSIC2LATENT). THE M2L “FINE-TUNE CONTROL” IS THE CONTINUED-TRAINING BASELINE WITH MIXING DISABLED. THESE QUANTITIES ARE INVARIANT TO THE INTERVENTION OF SECTION III.

decode-vs-decode by +4.06 dB. On the test set, $\mathcal{L}_{mix}^{dec}$ improves ground-truth equivariance from 8.0 to 10.2 dB and is the only configuration with MixRate < 1; reconstruction SI-SDR spans 11.2–11.4 dB and FAD 0.044–0.046, so we observe no reconstruction cost at this rate, on single runs. Trained from scratch at 15.3× against a matched control, the combined $\mathcal{L}_{mix}^{dec}$ +disc recipe improves SI-SDR_{lin}^{gt} from 6.8 to 8.6 dB and MixRate from 1.19 to 1.05 at equal reconstruction, while FAD rises from 0.052 to 0.061. That pair differs by the whole recipe, so it does not isolate $\mathcal{L}_{mix}^{dec}$ at that rate.

Level, not only shape. The gain SI-SDR discards: the decoded mix is 1.6 dB too quiet without the loss and 0.9 dB with it, against a reconstruction level error of 0.8 dB.

The mixed-path discriminator buys the metric, not the property. At 7.66×, adding an adversarial term on the mixed path leaves ground-truth equivariance unchanged (10.2 → 9.9 dB, MixRate 0.97 → 1.05) while raising decode-vs-decode SI-SDR_{lin} from 12.1 to 16.0 dB. The adversarial term increases decode-consistency without improving the latent-interpolation output’s ground-truth SI-SDR, so the two metrics rank these configurations differently. We do not identify the mechanism, and we never score the mixed-input reconstruction route against ground truth. Whether the discriminator acts alone is untested.

Architecture. The same mixed-latent supervision on Mu-

sic2Latent [11], under its native consistency loss, improves ground-truth equivariance against a continued-training control (−9.5 → −7.8 dB) while waveform agreement stays poor (−11.2 dB subtraction). The systems differ in too many respects to isolate a cause; we claim nothing about consistency autoencoders in general.

C. Decode-vs-decode scores are protocol-sensitive

For stochastic decoders the decode-vs-decode metric is confounded a second way. A consistency-model decoder [11] draws fresh noise per call: decoding the *same* latent twice is deterministic under shared noise but scores −1.8 dB under independent noise, with no latent arithmetic at all. On identical pairs the same checkpoints score −9.1 and −7.3 dB when the two decodes draw independent noise and +3.6 and +5.9 dB when they share it: a 13 dB swing driven by the evaluation protocol rather than the model. Which protocol prior work used is not stated in enough detail for us to determine. We use shared decode noise everywhere.

VI. CONCLUSION

Latent stem subtraction is measured in coordinates that the model never fixes. Recentering the latent space at the encoded silence leaves the autoencoding function and every unit-sum combination pointwise unchanged, for arbitrary nonlinear encoders and decoders, yet it moves subtraction scores by several decibels and, on five fixed checkpoints, reduces the advantage that mixing supervision appeared to have there by 87–95% on a pre-registered 240-track held-out corpus, leaving a residual 0.12 to 0.42 dB. What the supervision buys is convex equivariance, which the intervention cannot affect and which improves at both rates. Latent arithmetic should therefore be reported against a declared origin and against the model’s own reconstruction of the target. **Limitations.** Comparisons use single training runs: fine-tunes share a starting checkpoint, the from-scratch pair only a configuration, as training sets no explicit seed. Intervals quantify recording-level evaluation variation conditional on these checkpoints, not variation across training runs. Selection used the test split; recovery is exact only for an affine encoder; audio is mono and untested by listeners.

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